DO NOW — RECALL

Programming Techniques · Functions and Procedures

Name: _____ Class: _____ Date: _____

1. What is the final result of x=5; x-=2;?

2. What does a binary search need first?

3. How many times runs for(i=0;i<4;i++)?

4. What does a truth table list?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

3

2. Model answer

Sorted input

3. Model answer

4

4. Model answer

Outputs for input combinations

By the end, you can…

• Explain returned values and void methods

• Trace parameter values

• Define and call a C# method

Vocabulary — know these terms

Recall the term — function

Recall the term — procedure

Recall the term — parameter

Recall the term — argument

Recall the term — return

NEW LEARNING

Read, annotate and explain

Return a computed value

static int Double(int value) { return value * 2; } int result = Double(6); Console.WriteLine(result);

Explain the important decision

The argument is copied into a local parameter. return sends a computed value to the caller; it does not itself print the value.

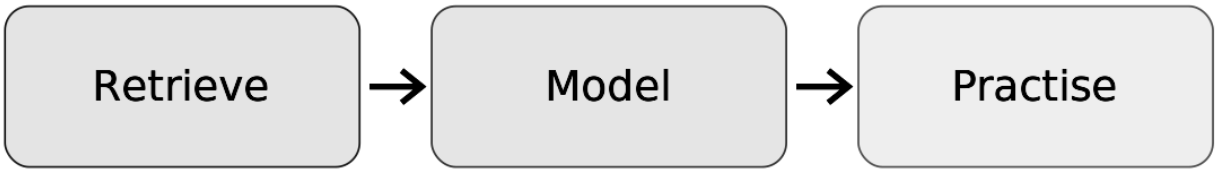
Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Return a computed value

static int Double(int value) { return value * 2; } int result = Double(6); Console.WriteLine(result);



Three-stage diagram showing retrieve, model, and practise

I DO — WORKED EXAMPLE

Follow the worked method

Return a computed value

Pass the argument

Double(6) starts a call with local parameter value=6. The caller waits for an integer result.

Calculate and return

value*2 evaluates to 12. return 12 ends this call and supplies that integer to the caller.

Use the returned value

The assignment stores 12 in result; Console.WriteLine(result) prints it. A void procedure can print, but supplies no value for this assignment.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

1. Problem 1

A function adds a and b; what is Add(3, 4)?

Method / working:

2. Problem 2

A void method prints a greeting. Does it return a string?

Method / working:

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Problem 1 A function adds a and b; what is Add(3, 4)? Discuss, then respond

DISCUSS — Compare approaches, walk through one test input, and explain how you would detect or correct a mistake.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

Guided 2: Problem 2 A void method prints a greeting. Does it return a string?

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Apply independently
Write a C# method returning the cube of an integer. [4]

Question 2

Apply independently
Explain the purpose of a parameter. [2]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Apply independently
Double(n) returns n*2. Calculate Double(Double(2)) and show both returns. [2]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

True or false: a void C# method must return a calculated value.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Check your thinking

7.

IMPROVE: Explain one correction to your own reasoning.

Guided 2 — original CHECK

Discuss, then respond

A void method prints a greeting. Does it return a string?

Check your thinking

No: void has no return value.

IMPROVE: Explain one correction to your own reasoning.